# HLD of messaging APP ( whatsapp)

Requirements → Entities → API → Data-flow

→ HLD → Deep-Dive
↳ H·S
↳ QPS
↳ Scale

Requirements → ① functional Req ② Non functional Requirements

## functional Req

① USER A → USER B
send / Receive msgs.

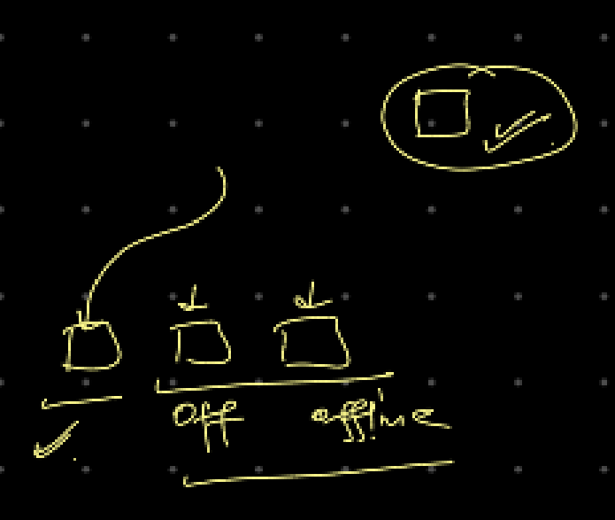
② Group chats →
→ I send → Grp → all members should receive msg
→ if I am part of a group → G receives msg → I get it.

③ offline → msg is lost (X)

④ ack, → ✓, ✓✓, ✓✓

⑤ support media → as an attachment

⑥ support multiple devices → clients


✓
off
offline

② Non-functional Requirements.
↳ Technical

Latency → 500 ms low

Guarantee delivery

→ Billion of user, high throughput →

→ msg are not stored permanently ⇒ dont store if all clients received it

→ fault tolerant → resilient

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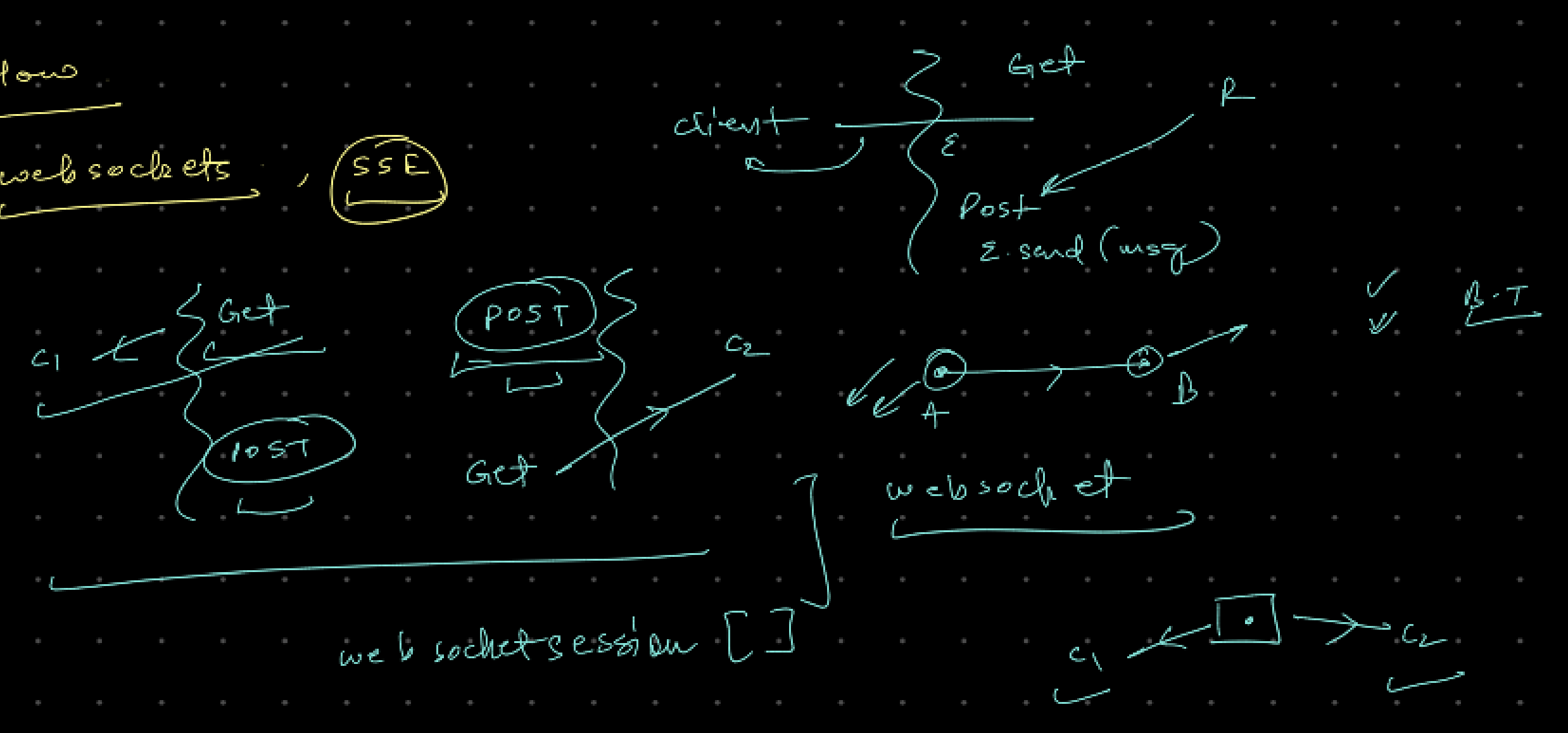
## Entities

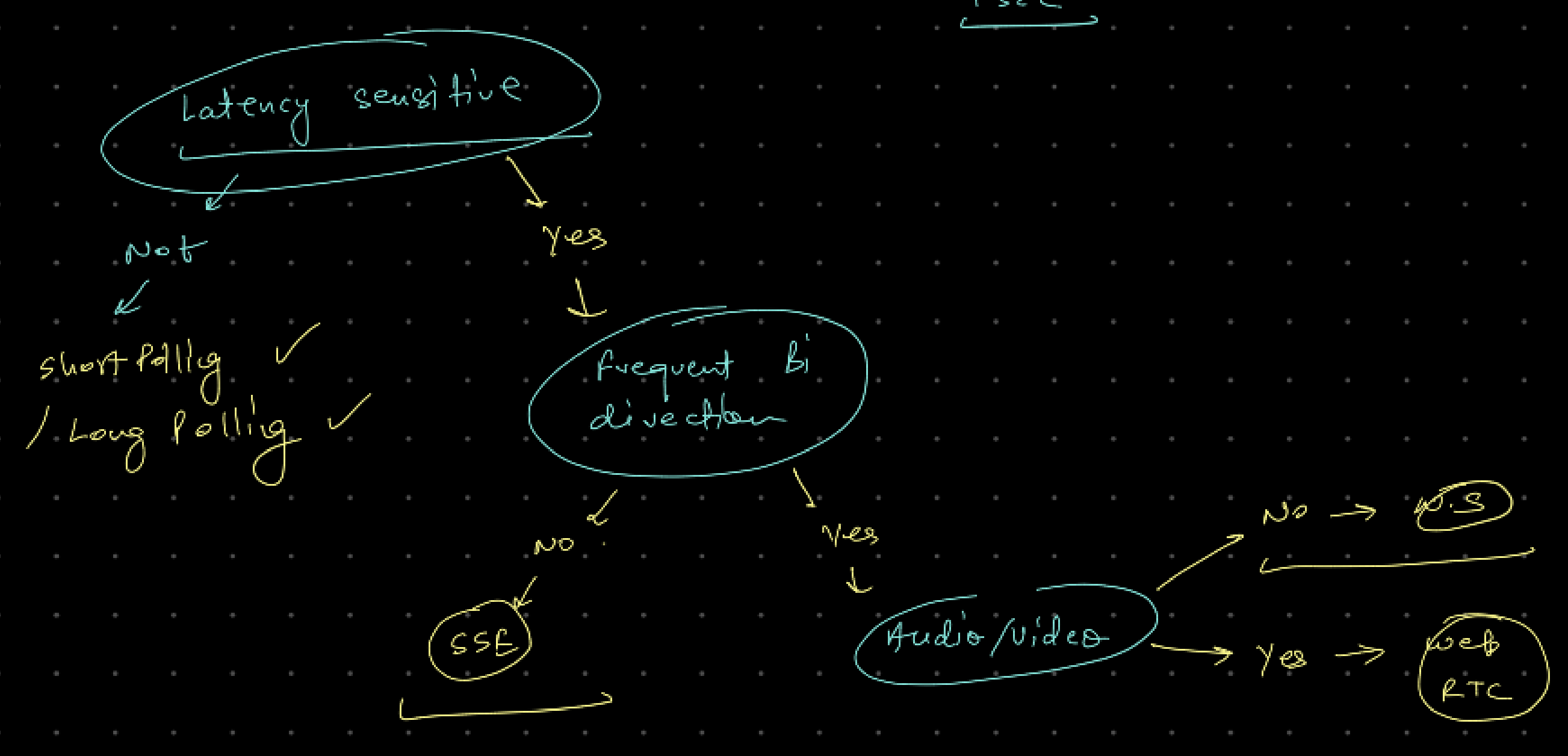
✓ user, client, chat, msg

— flow

websockets, SSE

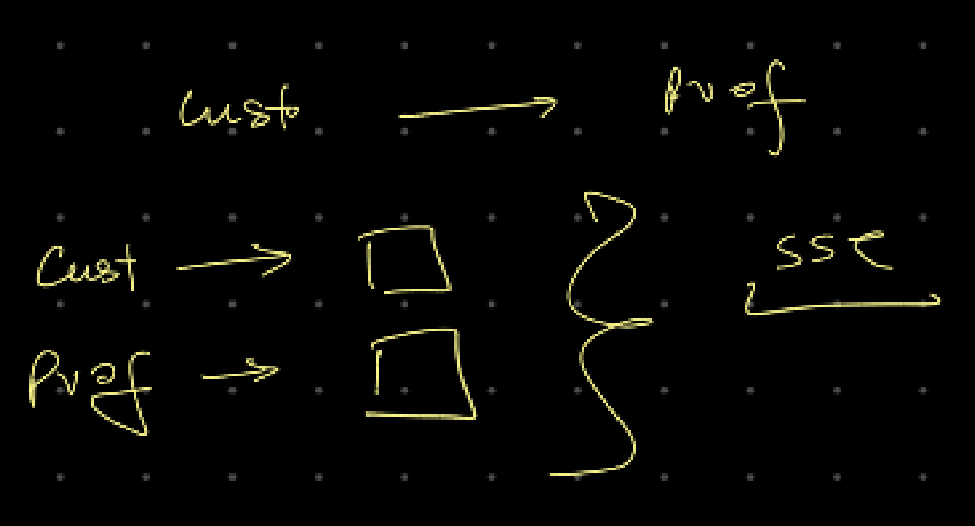

client
Get
Post
R
E
E.send (msg)
C1
Get
POST
POST
Get
C2
A
B
B.T
websocket
websocketsession [ ]
C1
C2

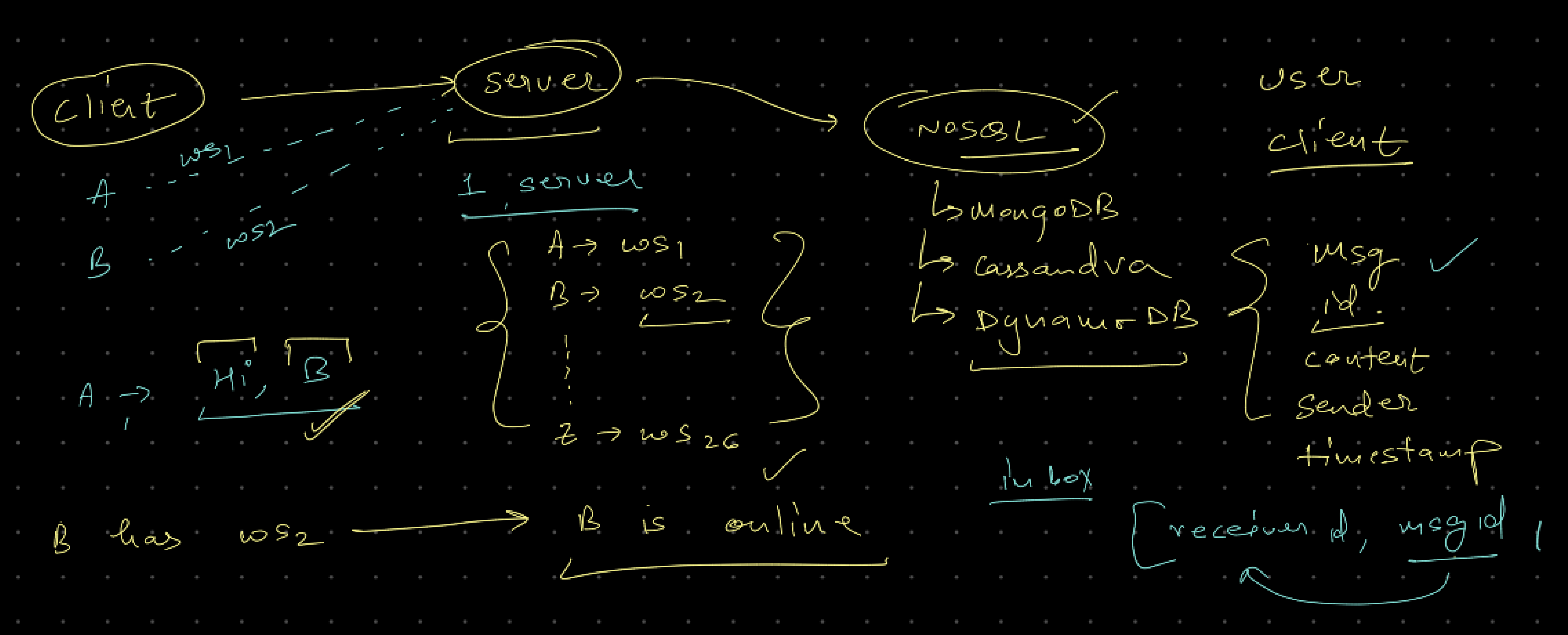
C1
C2
→ short polling
1 sec

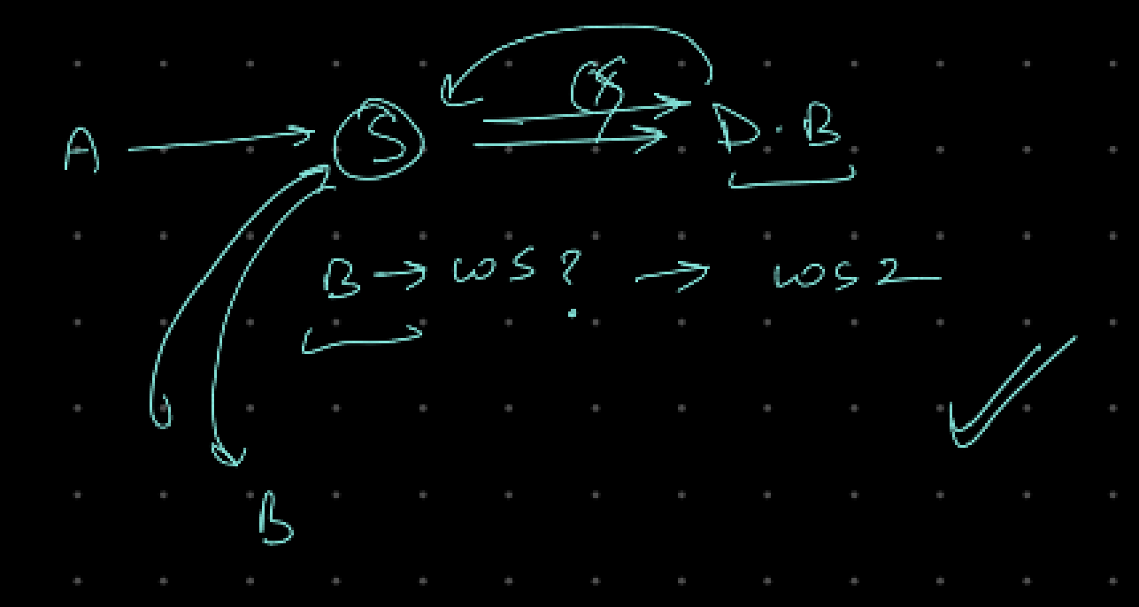

Latency sensitive
Not
Yes
short Polling ✓
/ Long Polling ✓
frequent Bi direction
No
Yes
SSE
Audio/Video
No → W.S
Yes → Web RTC

1 web ⇒ 30 KBs

16 GB RAM ⇒ 500K


cust
prof
Cust
Prof
SSE


Cliet
Server
NoSQL
↳ MongoDB
↳ Cassandra
↳ DynamoDB
user
client
msg ✓
id
content
sender
timestamp
A
B
ws1
ws2
1 server
A → ws1
B → ws2
Z → ws26
A → Hi, B
B has ws2
B is online
Inbox
[receiver id, msg id

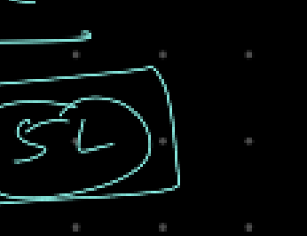

A
S
D.B
B → ws? → ws2
B

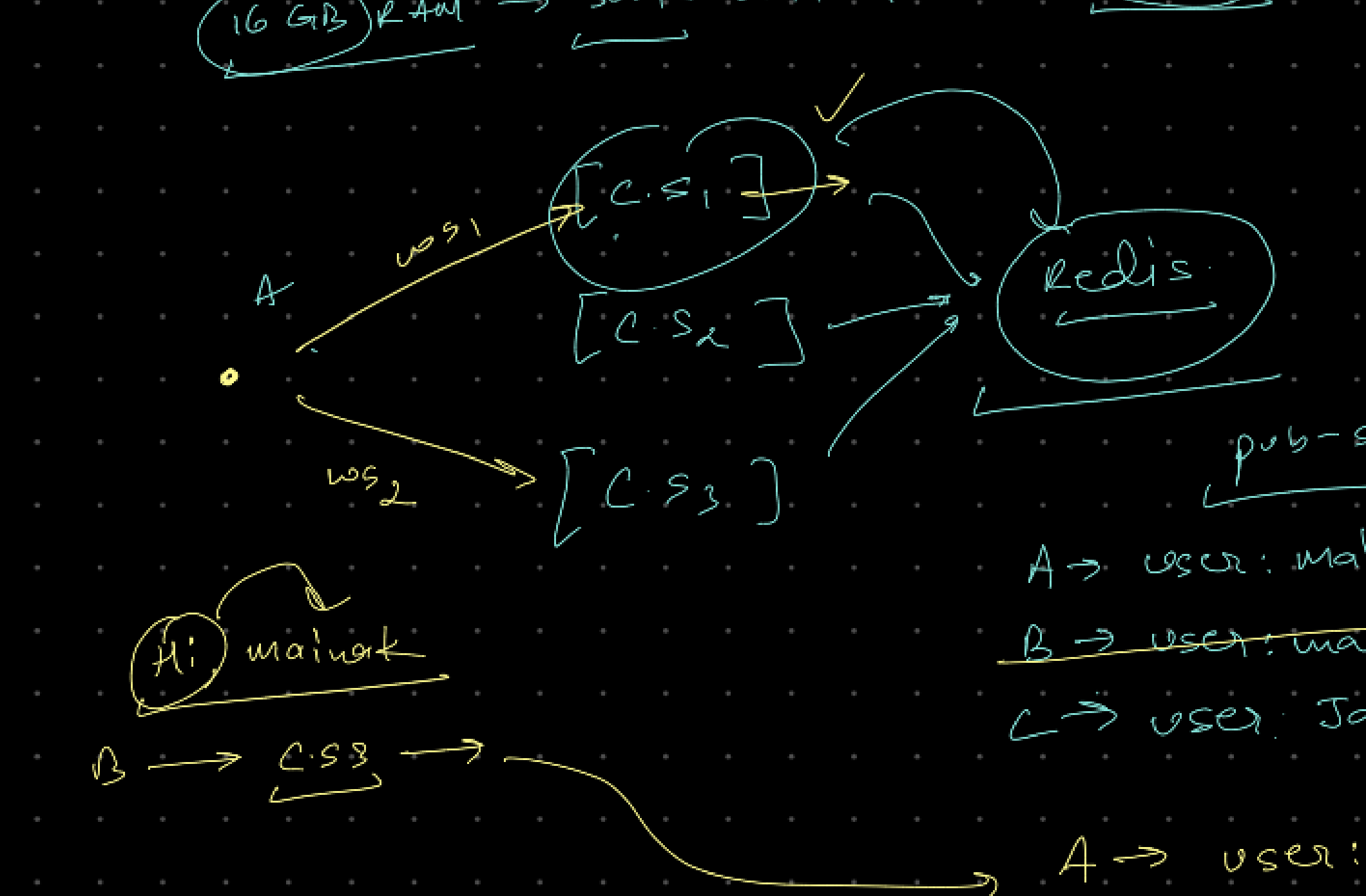
B can be offline

16 GB RAM → 500K connections → 20 KB

1B
↳ 1000 M

10% ⇒ 100M
online.




C.S1
C.S2
C.S3
Redis
A
ws1
ws2
Hi mainak
B → C.S3 →

pub-sub → channels.

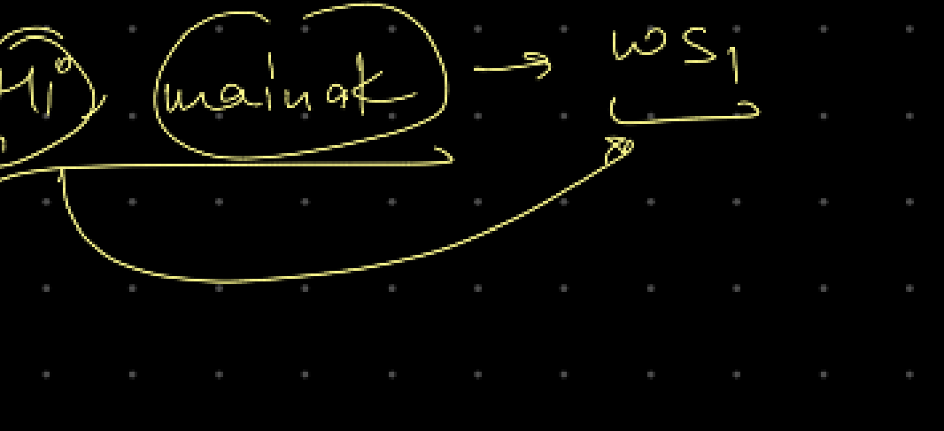
A → user: mainak : ~~laptop~~ ✓ ← S1

~~B → user: mainak : phone~~

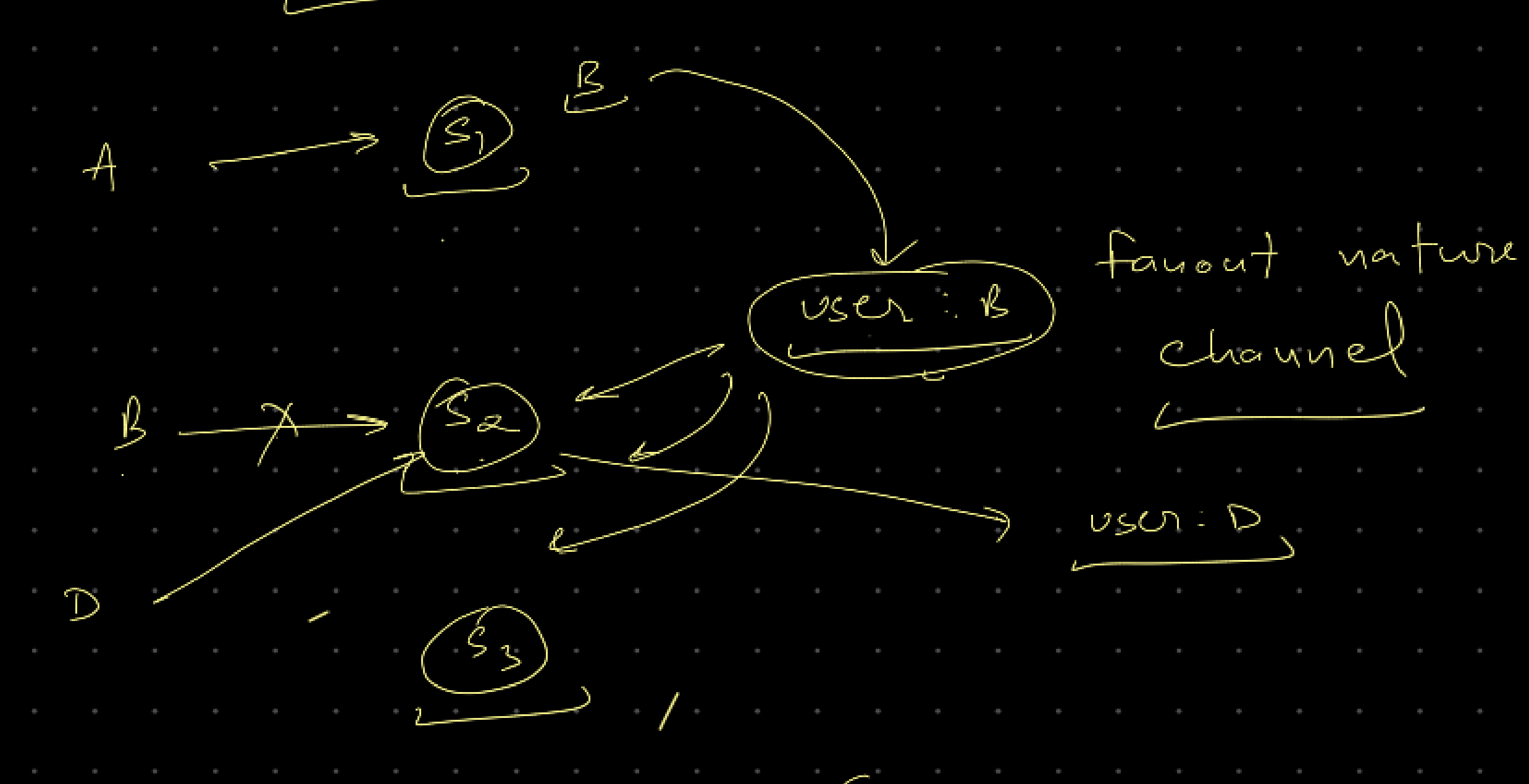
C → user: Jaikant : phone

A → user: mainak

msg

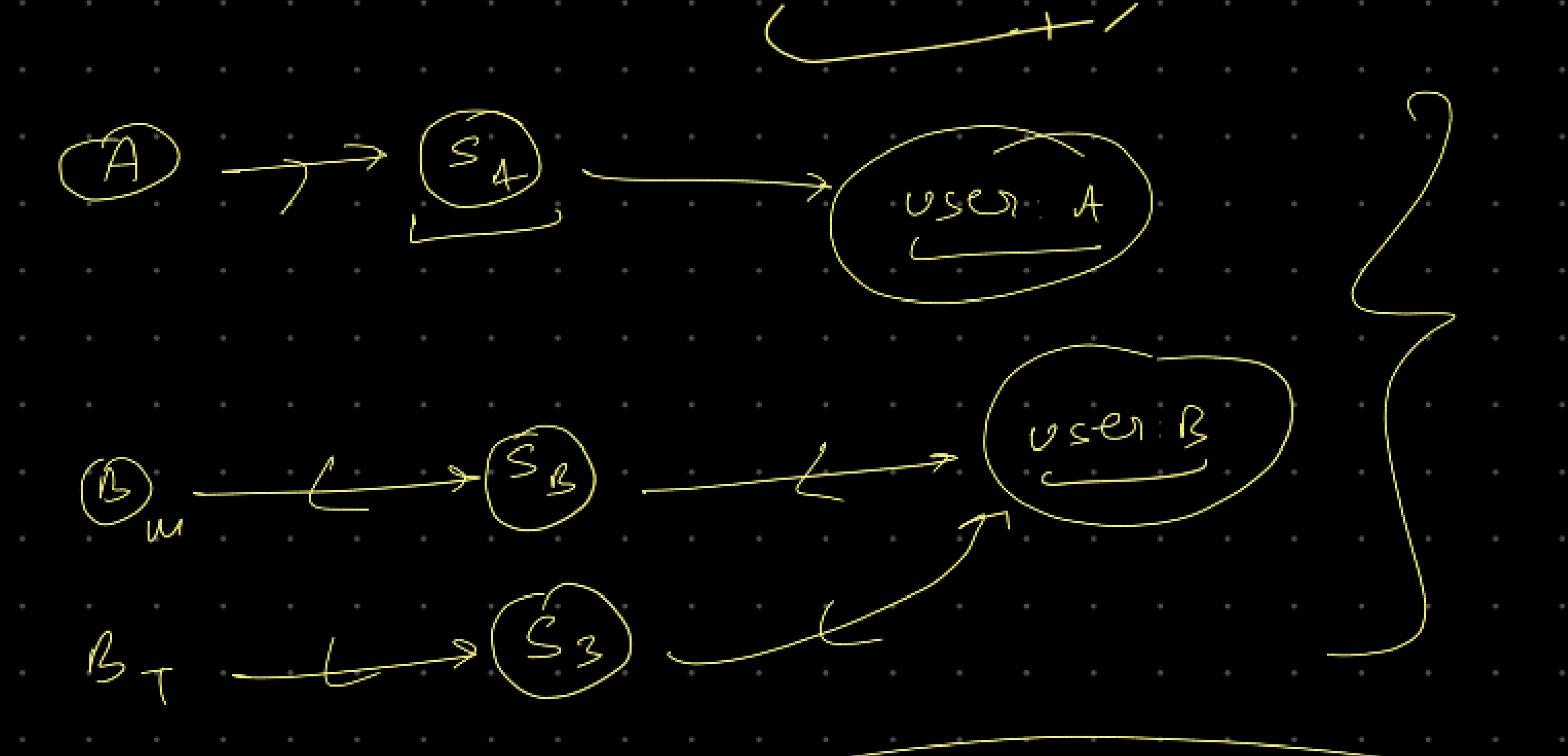

Hi
mainak
→ ws1

string → [ Each online user → channel ]

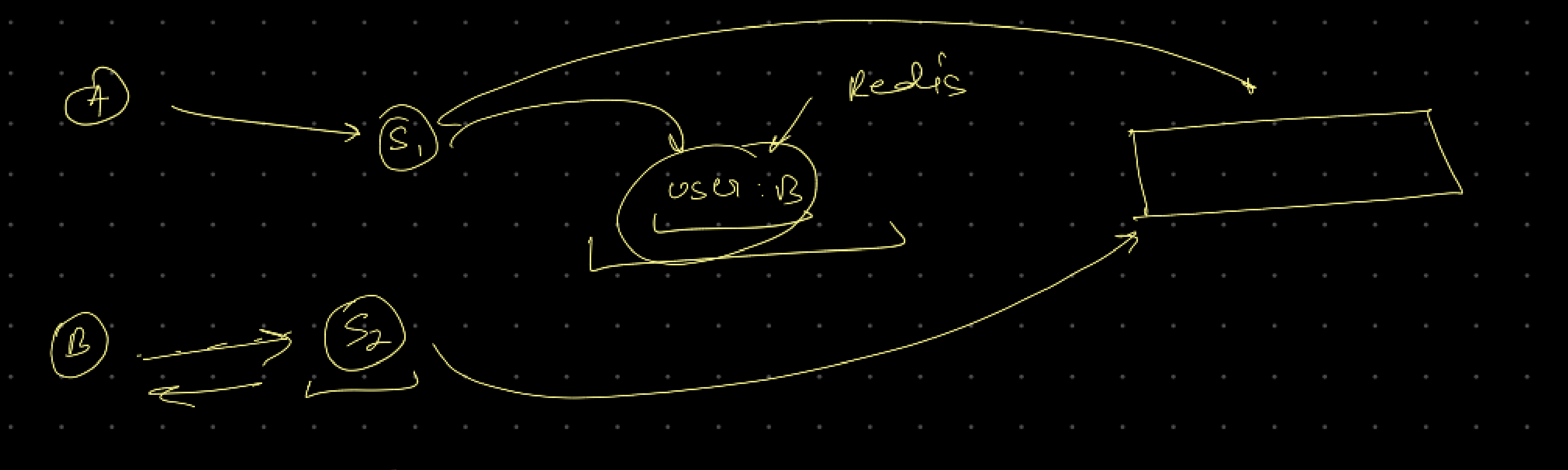

A
S1
B
user : B
B
S2
D
user : D
S3

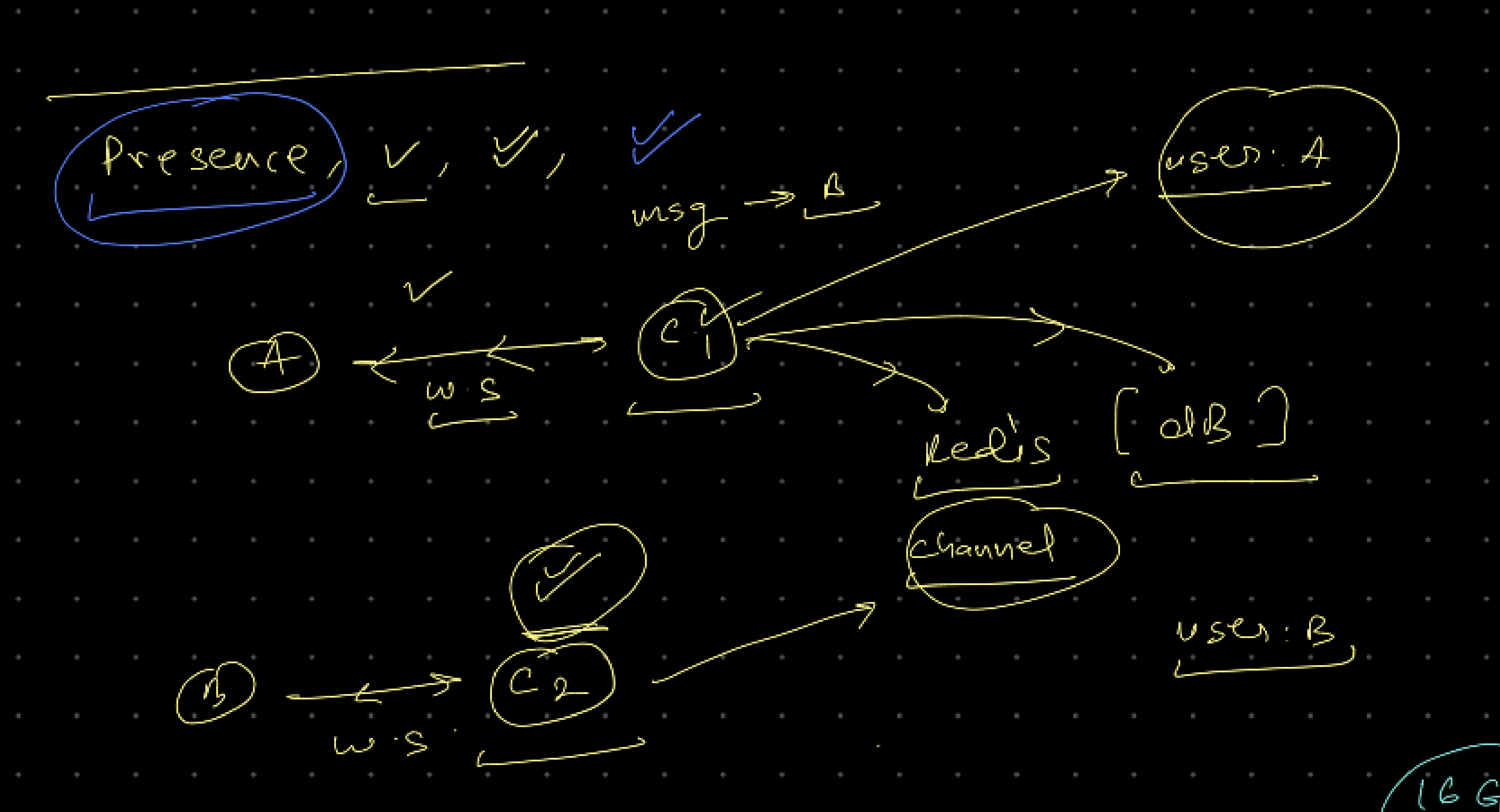
fanout nature of channel

S5 S6 S7


A
S4
user : A
B_M
S_B
user : B
B_T
S3

fire & forget


A
S1
Redis
user : B
B
S2


Presence
✓, ✓✓, ✓✓
msg → B
user : A
A
w.s
C1
Redis
[ dB ]
channel
B
w.s
C2
user : B

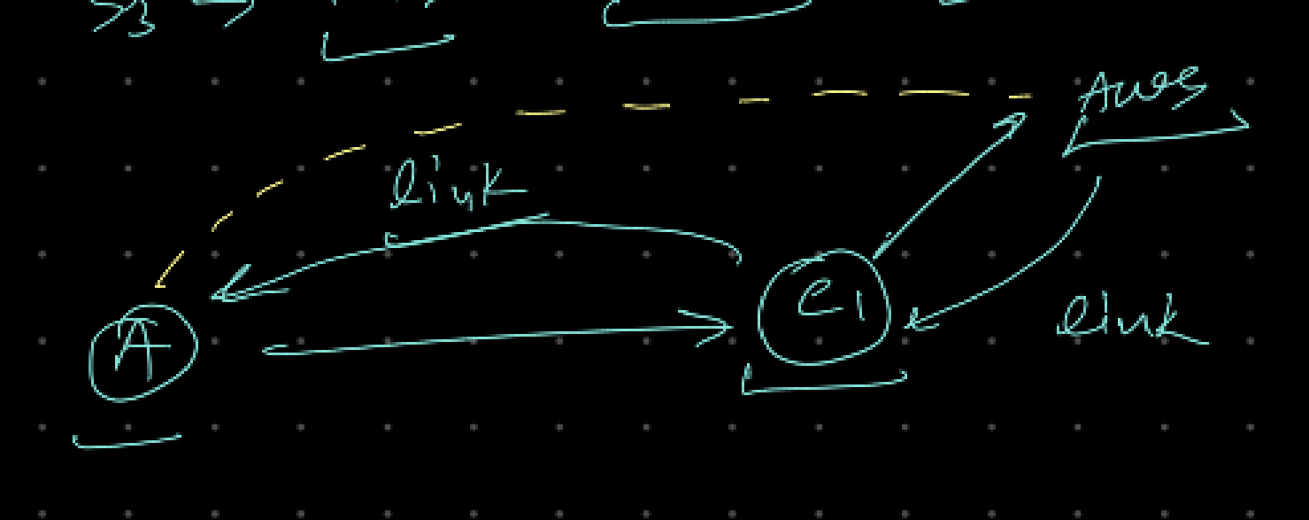
16 GB → 1GB
500K ✓

attachment → images, videos, [X]
1GB

AWS → S3, Azure → Blob storage

GCP → Blob - storage

S3 → 1GB → link → pre-signed


AWS
link
A
C1
link